

Yuchan (Carlos) Choi

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EDUCATION

Northwestern University, Evanston, Illinois
BS Mechanical Engineering

Expected Jun 2027
GPA: 3.75 / 4.00

EXPERIENCE

Vaccine Manufacturing Packaging Operations Intern | Merck | West Point, PA

Jun 2026 – Sep 2026

- Analyzed packaging-line data to trend production losses in a GMP-controlled space; supported 350K+ vaccine doses/week
- Designed and 3D printed custom storage fixtures using mathematical optimization across 6+ design iterations to close CAPAs on storage cross-contamination and increase capacity from 720 to 800 tubes; incorporated cross-functional operator feedback
- Developed Python/Jupyter, SQL, JavaScript & GitHub tools to document legacy code for automated line-clearance systems
- Built EBR/SAP troubleshooting logs with Automation & Quality to trend equipment issues and accelerate root-cause resolution

Undergraduate Robotics Researcher | HAND Engineering Research Center | Evanston, IL

Feb 2025 – Present

- Designed 10+ SOLIDWORKS/3D-printed prototypes for a cable-driven 1-DOF teleoperation robot, optimizing mass, impedance, and manufacturability; supporting research into improved transparency in bilateral controls through hybrid actuation
- Integrated ADS1220 ADCs for strain-gauge acquisition using SPI and custom C++ libraries, increasing sampling rate by 4×
- Designed KiCad PCBs integrating force sensing, signal conditioning, filtering, and power-protection circuitry for tendon robot

R&D Engineering Intern | Hendrickson – Advanced Technology Group | Woodridge, IL

Jul 2025 – Sep 2025

- Designed 50+ NX CAD models and 3D-printed prototypes to validate mechanical integration of PCBs, hydraulics, and sensors
- Optimized BLE sensor gain and placement using packet-sniffing analysis, reducing sensor data loss by 30% on ADC channels
- Researched RF-based tire localization methods for IoT-enabled suspension systems and new product development

Quality Assurance & Certifications Intern | Angel Robotics | Seoul, Korea

Jul 2024 – Aug 2024

- Documented wearable robotics research papers, patents, and international medical device regulations (ISO 13485, FDA 510(k), CE); aided the Technical Sales and Quality teams in the analysis of global market trends, and adherence to GMP and regulations
- Revised 300+ pages of technical and clinical evaluation reports under GDP for international medical-device certification

PROJECTS

Northwestern Solar Car 8 (SC8) | Mechanical Team Member – Chassis, Ergonomics, Manufacturing Training

- Developed 30+ SOLIDWORKS designs and prototypes to iterate and validate fit of critical vehicle components
- Reduced chassis frame-arm misalignment by 20% by leading a team of 4 to design and fabricate steel extension sleeves using CNC machining, correcting welding gaps and accelerating assembly by 3 days
- Led hands-on trainings for 25+ new members on mill, lathe, and fabrication equipment, enforcing safe manufacturing practices

65 kW Multi-Stage Transmission Design | Project Co-Lead – 3rd Shaft Principal Engineer

- Designed 3-shaft/4-gear transmission system capable of reducing from 3000 to 365 RPM with 1.51% error; optimized gear geometry and mechanical interfaces utilizing Onshape; received highest rated project score on Mechanical Design Course
- Sized gears, shafts, keys & SKF bearings using AGMA, fatigue/yield & L10 life analyses; developed tolerance-controlled assemblies, engineering drawings, and a comprehensive final report on manufacturability and maintenance of the final design

Electronic Fidget Spinner | Project Lead – SHPE Community Outreach Committee

- Spearheaded development of LED-embedded smart fidget spinners to support Northwestern's Society of Hispanic Professional Engineers' (SHPE) community outreach initiatives in collaboration with Electrical Engineering faculty
- Designed custom PCBs, soldered electrical components, and iterated multiple 3D printed mounts for optimized weight distribution and manufacturability; made 40 fidget spinners to help promote SHPE to local high school students

SKILLS

Technical: AutoCAD, SOLIDWORKS, Siemens NX, Onshape, MATLAB, Simulink, Python, C, C++, Arduino, KiCad

Language: English, Korean, Spanish